



**COMPETENCY STANDARD
FOR
INDUSTRIAL ELECTRICAL AND ELECTRONIC
TECHNOLOGY FOR TEXTILE INDUSTRY
(TEXTILE SECTOR)**

**Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance
People's Republic of Bangladesh**

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The Competency Standard for Industrial Electrical and Electronic Technology for Textile Industry is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry so that individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

This document has been developed for use in training under the Skills for Industry Competitiveness and Innovation Program (SICIP) to meet the skills requirements of the Textile sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The textile and apparel sector is the largest manufacturing and export sector of Bangladesh. Modern spinning, weaving, knitting, dyeing-finishing and garment factories depend on highly automated machinery driven by induction and servo motors, variable frequency drives, programmable logic controllers, sensors and electronic control systems, supported by substantial utility infrastructure such as compressors, humidification plants, boilers, generators and effluent treatment plants. Reliable operation and rapid troubleshooting of these electrical and electronic systems directly determine machine uptime, product quality and energy cost. The industry therefore requires technicians who combine a diploma-level engineering foundation with hands-on, textile-specific competence in industrial electrical and electronic technology.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of diploma engineering graduates by implementing market-responsive and job-focused training programs through public and private training providers. This requires the development of competency standards for each of the occupations/trades, which provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical sessions, which aligns with modern learning preferences and supports flipped classroom models. Industry visits and industry-based practical sessions in textile mills are strongly recommended to expose trainees to actual production machinery.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements. The document details the format, sequencing, wording, and layout of the Competency Standard for the occupation, which comprises Units of Competence and their corresponding Elements.

OVERVIEW

A Competency Standard is a written specification of the knowledge, skills, and attitudes required for the effective performance of an occupation or trade role, corresponding to the standards of performance required in the workplace.

A Competency Standard:

- Provides a consistent and reliable set of components for training, recognizing, and assessing workers' skills
- Enables industry-recognized qualifications to be awarded through direct assessment of practical workplace competencies
- Encourages the development and delivery of flexible training that suits individual needs and industry requirements
- Supports learning and assessment in a work-related environment leading to verifiable workplace outcomes

Competency Standards describe the skills, knowledge, and attitudes needed to perform effectively in the workplace. They acknowledge that workers can achieve vocational and technical competency through many pathways, emphasizing what the learner can DO rather than how or where they learned it.

This Competency Standard for Industrial Electrical and Electronic Technology for Textile Industry has been developed with the following key considerations:

- Target Trainees: Graduates of Diploma in Engineering (Electrical, Electronics, Mechatronics, Power, Instrumentation, Textile or related technologies).
- Job Roles: Electrical/Electronic Maintenance Technician, Junior Electrical Engineer, Automation Technician, Utility Electrical Technician and Machine Electrician in spinning, weaving, knitting, dyeing-finishing and garment factories.
- Industry Context: Bangladesh's textile industry including spinning mills, weaving mills, knitting units, wet processing (dyeing, printing and finishing) plants, garment factories and their utility sections.
- Course Duration: 360 hours delivered over three months.
- Batch Size: 25 trainees per batch; a total of 200 trainees (8 batches) will be trained under this course.
- Language of Delivery: Training may be delivered in Bangla with English technical terminology; practical demonstration and oral assessment methods are prioritized.
- Delivery Mode: Approximately 30% theory and 70% practical, including industry exposure.

A Unit of Competency describes a distinct, work-related activity normally undertaken by one person in accordance with industry standards. Each unit is documented in a standard format comprising:

- Unit Code, Title, and Descriptor
- Elements of Competency and Performance Criteria
- Range of Variables
- Curricular Content Guide
- Assessment Evidence Guide

Unit codes follow the format SICIP-BPI-IEET-nn-X, where IEET denotes Industrial Electrical and Electronic Technology and X denotes G (Generic), S (Sector Specific) or O (Occupation Specific).

UNITS & ELEMENTS AT A GLANCE

Category of Competency	Duration
Generic Competencies	30 Hrs.
Sector Specific Competencies	10 Hrs.
Occupation Specific Competencies	320 Hrs.
Total	360 Hrs.

Generic Competencies (30 Hrs.)

Code	Unit of Competency	Elements of Competency	Hours
SICIP-TEX-IEET-01-G	Apply Occupational Health and Safety (OHS) Practices at Textile Worksites	1. Identify OHS policies and procedures 2. Use personal health and safety practices 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-TEX-IEET-02-G	Operate in a Team Environment	1. Identify own role and responsibilities within the team 2. Support and cooperate with team members 3. Communicate and share information within the team 4. Contribute to problem-solving within the team	10
Total Hours			20 Hrs.

Sector Specific Competencies (10 Hrs.)

Code	Unit of Competency	Elements of Competency	Hours
SICIP-TEX-IEET-01-S	Interpret Textile Manufacturing Processes and Electrical Systems in Textile Mills	1. Identify textile manufacturing processes and machinery 2. Identify electrical and electronic systems in textile mills 3. Recognize textile-specific environmental conditions affecting electrical equipment 4. Apply regulations and standards relevant to electrical work in textile mills	10
SICIP-TEX-IEET-02-S	Apply Green Practices in the Textile Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hours			20 Hrs.

Occupation Specific Competencies (320 Hrs.)

Code	Unit of Competency	Elements of Competency	Hours
SICIP-TEX-IEET-01-O	Apply Industrial Electrical Fundamentals and Interpret Electrical/Electronic Circuit Drawings	1. Interpret electrical and electronic drawings and documents 2. Analyze DC and AC circuits 3. Analyze three-phase systems and power 4. Test basic electronic components and circuits 5. Use electrical measuring and testing instruments	30
SICIP-TEX-IEET-02-O	Install and Maintain Industrial Power Distribution Systems in Textile Mills	1. Interpret power distribution requirements 2. Install LT panels and distribution boards 3. Install cables and cable trays 4. Install earthing and lightning protection 5. Test and commission power distribution installation 6. Perform preventive maintenance of switchgear and panels	40
SICIP-TEX-IEET-03-O	Install, operate and Maintain Industrial Motor Control Systems	1. Identify motors used in textile machinery 2. Select control and protective devices 3. Install and wire motor starters 4. Connect and test motors 5. Test motor control circuits 6. Maintain motors and starters	55

Code	Unit of Competency	Elements of Competency	Hours
SICIP-TEX-IEET-04-O	Install, Configure and Operate Variable Frequency Drives (VFD) and Soft Starters	1. Interpret VFD and soft starter principles and applications 2. Install and wire VFDs and soft starters 3. Configure drive parameters 4. Commission and operate drives 5. Troubleshoot and maintain drives	40
SICIP-TEX-IEET-05-O	Program and Operate PLC and HMI Systems for Textile Machinery Automation	1. Identify PLC hardware and architecture 2. Wire PLC inputs and outputs 3. Develop ladder logic programs 4. Configure HMI 5. Control VFD with PLC & Sensors 6. Commission PLC-based control of textile machine functions 7. Troubleshoot PLC systems and manage program backups	50
SICIP-TEX-IEET-06-O	Install, operate and Maintain Sensors, Instrumentation and Electronic Control Circuits	1. Identify sensors and transducers used in textile machinery 2. Install and calibrate sensors and transmitters 3. Assemble and test electronic control circuits 4. Maintain electronic control cards of textile machines 5. Test instrumentation loops	40

Code	Unit of Competency	Elements of Competency	Hours
SICIP-TEX-IEET-07-O	Diagnose and Troubleshoot Electrical and Electronic Faults in Textile Machinery	1. Collect fault information 2. Apply systematic troubleshooting procedures 3. Diagnose faults in power, control and electronic circuits 4. Rectify faults and replace components 5. Test and restore machine to operation 6. Record and report maintenance activities	50
SICIP-TEX-IEET-08 -O	Implement Energy Efficiency and Power Quality Management	1. Conduct energy audit of textile electrical loads 2. Improve power factor 3. Identify and mitigate power quality problems 4. Implement energy conservation measures 5. Monitor and report energy performance	15
Total Hours			320 Hrs.

COMPETENCY STANDARD: INDUSTRIAL ELECTRICAL AND ELECTRONIC TECHNOLOGY FOR TEXTILE INDUSTRY

The Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES AT TEXTILE WORKSITES	Nominal Duration: 10 hrs.	Unit Code: SICIP-TEX-IEET-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices at textile industry worksites. It specifically includes the tasks of identifying OHS policies and procedures, using personal health and safety practices, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
Identify OHS policies and procedures	1. <u>OHS policies</u> and safe operating procedures of the textile mill are interpreted. 2. Safety signs and symbols used on textile production floors and electrical rooms are identified and followed. 3. Emergency response, evacuation procedures and other contingency measures are interpreted correctly.
Use personal health and safety practices	1. <u>Personal protective equipment (PPE)</u> appropriate to electrical and textile work is selected, inspected and worn. 2. Common health issues in textile mills such as cotton dust exposure, noise and heat stress are recognized. 3. Safe manual handling, housekeeping and machine guarding practices are applied.
Report hazards and risks	1. <u>Hazards and risks</u> in electrical and textile work areas are identified. 2. Hazard and risk assessment and control measures are interpreted. 3. Hazards, near-misses and incidents are reported using workplace formats.
Respond to emergencies	1. Alarms and warning devices are responded to promptly. 2. <u>Emergency response plans and procedures</u> are followed. 3. <u>First aid procedures</u> including treatment for electric shock are identified.

Range of Variables:

Variables	Range (may include but not limited to)
OHS policies	- Organizational OHS policies - Bangladesh Labour Act and Labour Rules (OHS provisions) - International OHS requirements and buyer compliance codes - Fire safety rules and regulations
Personal protective equipment	- Safety glasses - Ear plugs/ear muffs - Insulated gloves - Dust mask/respirator - Helmet - Apron - Safety shoes with insulated soles
Hazards and risks	- Electrical shock and arc flash - Rotating machinery and nip points - Cotton dust and fibre fly - Noise - Heat and humidity - Chemical exposure in dyeing/finishing - Fire from lint accumulation
Emergency response plans and procedures	- Firefighting procedures - Earthquake response procedures - Electrical accident response - Chemical spill response - Medical and first aid
First aid procedures	- Isolating power before rescuing shock victim - Cardiopulmonary resuscitation (CPR) - Washing open wounds - Washing chemically affected areas - Applying bandage - Treating burns

Curricular Content Guide:

Underpinning Knowledge	- Workplace OHS policies and procedures - Work safety procedures in textile mills - Emergency response procedures: firefighting, earthquake response, accident response - Types of hazards (electrical, mechanical, physical, chemical, biological) and their effects - OHS awareness and safety signage - Personal protective equipment (PPE) - Electric shock treatment and CPR

Underpinning Skills	- Identifying OHS policies and procedures - Applying personal health and safety practices - Reporting hazards and risks - Responding to emergencies
Underpinning Attitudes	- Commitment to occupational health and safety - Promptness in carrying out activities - Sincere and honest to duties - Environmental concerns - Eagerness to learn - Tidiness and timeliness - Respect for rights of peers and seniors in the workplace - Communication with peers, subordinates and seniors in the workplace
Resource Implications	The following resources must be provided: - Workplace (simulated or actual) - Workplace documents, signs and symbols - Codes of conduct - Projector - Learning manual - PPE, first aid kit and fire extinguishers - Tools, equipment and facilities appropriate to the process or activities - Materials relevant to the proposed activity

Assessment Evidence Guide:

Critical Aspects of Competency	Assessment requires evidence that the candidate: - identified OHS policies and procedures - applied personal health and safety practices (including PPE) - reported hazards and risks - responded to emergencies
Methods of Assessment	Competency should be assessed by: - Written test - Practical demonstration - Oral questioning - Portfolio (optional)
Context of Assessment	- Competency assessment must be done in an assessment/training center or in an actual or simulated textile mill workplace after completion of the training. - Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE IN A TEAM ENVIRONMENT	Nominal Duration: 10 hrs.	Unit Code: SICIP-TEX-IEET-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the tasks of identifying own role and responsibilities within the team, supporting and cooperating with team members, communicating and sharing information within the team and contributing to problem-solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify own role and responsibilities within the team	1.1 Personal role and responsibilities are identified within the maintenance team. 1.2 Importance of relationships within and outside the team is interpreted. 1.3 Roles and objectives of the team are identified and interpreted. 1.4 Roles and responsibilities of <u>team members</u> and work processes are identified.
Support and cooperate with team members	1. Team members are supported by sharing information, resources and assistance as needed. 2. Tasks are carried out cooperatively to achieve common work objectives. 3. Clear and respectful communication is maintained with all team members. 4. Differences are handled constructively to maintain a positive working environment.
Communicate and share information within the team	1. Other teammates' tasks are identified and support is provided when requested. 2. <u>Sharing information</u> and expertise is practiced to solve problems and put team success first. 3. Views and opinions of other team members are interpreted and respected.
Contribute to problem-solving within the team	1. Problems faced at the individual and team level are identified with insight into root causes. 2. A range of solutions and courses of action are identified together with benefits, costs and risks. 3. Ideas of others are recognized and advice is sought from those who have solved similar problems. 4. Solutions are evaluated beyond the first obvious answer.

Range of Variables:

Variables	Range (may include but not limited to)
Team members	• Maintenance supervisors and engineers • Skilled electricians and electronic technicians • Machine operators and production staff • Co-workers and trainees • Utility and ETP staff • Representatives from other departments
Sharing information	• Agenda and minutes • Shift handover reports • Breakdown and incident reports • Machine manuals • Visual and graphic materials • Emails, SMS and messaging groups • Policies, procedures and standards • OHS information

Curricular Content Guide:

Underpinning Knowledge	• Team goals and work processes • Roles and responsibilities of the team • Problem identification and solving techniques • Importance of views and opinions of other team members
Underpinning Skills	• Identifying own role and responsibilities within team • Communicating and cooperating with team members • Demonstrating problem solving within the team
Underpinning Attitudes	• Commitment to occupational health and safety • Promptness in carrying out activities • Sincere and honest to duties • Environmental concerns • Eagerness to learn • Tidiness and timeliness • Respect for rights of peers and seniors in the workplace • Communication with peers, subordinates and seniors in the workplace
Resource Implications	The following resources must be provided: • Workplace (simulated or actual) • Workplace documents • Projector • Learning manual • Tools, equipment and facilities appropriate to the process or activities • Materials relevant to the proposed activity

Assessment Evidence Guide:

Critical Aspects of Competency	Assessment requires evidence that the candidate: • identified own role and responsibilities within team • communicated and cooperated with team members • demonstrated problem solving within the team
Methods of Assessment	Competency should be assessed by: • Written test • Practical demonstration • Oral questioning • Portfolio (optional)
Context of Assessment	• Competency assessment must be done in an assessment/training center or in an actual or simulated textile mill workplace after completion of the training. • Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: INTERPRET TEXTILE MANUFACTURING PROCESSES AND ELECTRICAL SYSTEMS IN TEXTILE MILLS	Nominal Duration: 10 hrs.	Unit Code: SICIP-TEX-IEET-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret textile manufacturing processes and the electrical and electronic systems that support them. It specifically includes the tasks of identifying textile manufacturing processes and machinery, identifying electrical and electronic systems in textile mills, recognizing textile-specific environmental conditions affecting electrical equipment and applying regulations and standards relevant to electrical work in textile mills.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
Identify textile manufacturing processes and machinery	1. Textile manufacturing processes from fibre to finished fabric and garment are identified. 2. Major <u>textile machinery</u> at each stage of production and their functions are described. 3. Production flow and interdependence of departments are interpreted. 4. Impact of electrical downtime on production output and quality is explained.
Identify electrical and electronic systems in textile mills	1. Power supply arrangements including utility supply, captive generation and backup systems are identified. 2. <u>Electrical and electronic systems</u> of textile machinery are identified. 3. Utility services supporting production are identified and their electrical loads described.
Recognize textile-specific environmental conditions affecting electrical equipment	1. Environmental conditions such as fibre fly, lint, dust, humidity, heat and static electricity are recognized. 2. Effects of environmental conditions on motors, drives, panels and electronic cards are explained. 3. Suitable <u>ingress protection (IP) ratings</u> and enclosure types for different textile areas are identified. 4. Cleaning and cooling arrangements for electrical equipment in textile areas are described.

Elements of Competency	Performance Criteria
Apply regulations and standards relevant to electrical work in textile mills	1. Regulations and standards relevant to electrical work in textile mills are identified. 2. Buyer compliance and certification requirements related to electrical safety are recognized. 3. Roles of maintenance, production, utility and compliance departments are interpreted.

Range of Variables:

Variables	Range (may include but not limited to)
Textile machinery	• Blow room lines, carding machines, draw frames, combers, speed frames, ring frames, rotor spinning machines, autoconers • Warping, sizing machines, air-jet/rapier/projectile looms • Circular knitting and flat knitting machines • Jet dyeing machines, jiggers, stenters, dryers, compactors, printing machines • Cutting machines, industrial sewing machines, fusing and pressing machines
Electrical and electronic systems	• Main motor and inverter drives • Servo and stepper motor systems • PLC/microcontroller-based machine controllers • HMI and touch panels • Sensors (yarn break, stop motion, temperature, pressure, speed) • Heating elements and thermostats • Machine lighting and electronic monitoring systems
Ingress protection (IP) ratings	• IP20 for indoor clean panels • IP54/IP55 for dusty areas • IP65/IP66 for wet processing areas • Totally enclosed fan cooled (TEFC) motors
Regulations and standards	• Bangladesh Electricity Act 2018 and Electricity Rules • Bangladesh National Building Code (BNBC) electrical provisions • Bangladesh Labour Act 2006 and Labour Rules 2015 • Fire Prevention and Extinction Act • IEC standards for industrial installations and machinery safety • Buyer/brand compliance requirements and social/technical audits

Curricular Content Guide:

Underpinning Knowledge	• Textile production stages and flow • Functions of major textile machinery • Electrical and electronic systems of textile machinery • Utility services in textile mills and their electrical loads • Effects of dust, humidity, heat and static electricity on electrical equipment • IP ratings and enclosure selection • National regulations, standards and compliance requirements
Underpinning Skills	• Identifying textile manufacturing processes and machinery • Identifying electrical and electronic systems in textile mills • Reading and interpreting nameplate data • Recognizing environmental conditions and selecting suitable equipment protection • Applying regulations and standards
Underpinning Attitudes	• Commitment to occupational health and safety • Promptness in carrying out activities • Sincere and honest to duties • Environmental concerns • Eagerness to learn • Tidiness and timeliness • Respect for rights of peers and seniors in the workplace • Communication with peers, subordinates and seniors in the workplace
Resource Implications	The following resources must be provided: • Workplace (simulated or actual) • Charts, videos and models of textile processes and machinery • Machine manuals and nameplates • Copies of relevant regulations and standards • Projector and learning manual • Industry visit to a textile mill (recommended)

Assessment Evidence Guide:

Critical Aspects of Competency	Assessment requires evidence that the candidate: • identified textile manufacturing processes and machinery • identified electrical and electronic systems in textile mills • recognized textile-specific environmental conditions affecting electrical equipment • applied regulations and standards relevant to electrical work in textile mills

Methods of Assessment	Competency should be assessed by: • Written test • Practical demonstration • Oral questioning • Portfolio (optional)
Context of Assessment	• Competency assessment must be done in an assessment/ training center or in an actual or simulated textile mill workplace after completion of the training. • Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation Specific Competencies

Unit of Competency: APPLY INDUSTRIAL ELECTRICAL FUNDAMENTALS AND INTERPRET ELECTRICAL/ELECTRONIC CIRCUIT DRAWINGS	Nominal Duration: 30 hrs.	Unit Code: SICIP-TEX-IEET-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply industrial electrical fundamentals and interpret electrical and electronic drawings. It specifically includes the tasks of interpreting electrical and electronic drawings and documents, analyzing DC and AC circuits, analyzing three-phase systems and power, testing basic electronic components and circuits and using electrical measuring and testing instruments.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
Interpret electrical and electronic drawings and documents	1. <u>Electrical symbols</u> used in industrial circuit diagrams are identified. 2. <u>Electrical drawings</u> of textile machinery and mill installations are interpreted. 3. Machine manuals, wiring diagrams and terminal plans are used to locate components and connections. 4. Simple circuit diagrams and panel layouts are drawn using standard symbols.
Analyze DC and AC circuits	1. Ohm's Law and Kirchhoff's laws are applied to series, parallel and combined circuits. 2. <u>Circuit theorems</u> are applied to solve industrial circuit problems. 3. RL, RC and RLC circuits are constructed and their behaviour is measured and interpreted. 4. Power, power factor and energy in single-phase AC circuits are calculated and verified.
Analyze three-phase systems and power	1. Star and delta connections are identified and line/phase relationships are calculated. 2. Three-phase power for balanced and unbalanced loads is measured and calculated. 3. Phase sequence and voltage unbalance are measured and interpreted. 4. Load current, cable size and protective device rating for textile machine loads are estimated.

Elements of Competency	Performance Criteria
Test basic electronic components and circuits	1. Electronic components are identified and tested with a multimeter. 2. Rectifier, filter and regulated DC power supply circuits are assembled and tested. 3. Transistor switching, opto-coupler and relay logic circuits are assembled and tested. 4. Basic digital logic and timer circuits are tested.
Use electrical measuring and testing instruments	1. Measuring and testing instruments are selected according to the quantity to be measured. 2. Instruments are connected and readings are taken safely and accurately. 3. Oscilloscope is used to observe waveforms of AC, rectified and pulse signals. 4. Instruments are calibrated, handled and stored as per manufacturer instructions.

Range of Variables:

Variables	Range (may include but not limited to)
Electrical symbols	• Sources (AC/DC) • Resistor, capacitor, inductor, transformer • Switches, push buttons, limit switches • Relay coils and contacts, magnetic contactors • Circuit breakers, fuses, overload relays • Timers and counters • Motors, drives, sensors • PLC I/O symbols
Electrical drawings	• Single-line diagrams (SLD) • Schematic/control circuit diagrams • Power circuit diagrams • Wiring and terminal diagrams • Panel layout and general arrangement drawings • Cable schedules
Circuit theorems	• Kirchhoff's voltage and current laws • Thevenin's and Norton's theorems • Superposition theorem • Maximum power transfer theorem
Electronic components	• Resistors, capacitors, inductors • Diodes, Zener diodes, LEDs • Bipolar junction transistors, MOSFETs • SCR, TRIAC, IGBT • Opto-couplers • Voltage regulator ICs • Logic gates and timer ICs (555)

Variables	Range (may include but not limited to)
Relay	• Thermal overload relay • Solid state relay • Electro-mechanical relay
Measuring and testing instruments	• Digital multimeter • Clamp meter • Insulation resistance tester (Megger) • Earth resistance tester • Phase sequence tester • Power/energy meter and power quality analyzer • Tachometer • Digital oscilloscope and function generator • Thermal imaging camera/IR thermometer

Curricular Content Guide:

Underpinning Knowledge	• Electrical symbols, drawing conventions and drawing types • DC and AC circuit laws and theorems • Single-phase and three-phase power concepts, power factor • Star and delta connections • Cable sizing and protective device rating basics • Semiconductor devices and power supply circuits • Digital logic and timer basics • Principles and safe use of measuring instruments
Underpinning Skills	• Reading and drawing electrical and electronic circuit diagrams • Constructing and testing DC and AC circuits • Measuring three-phase voltage, current, power and phase sequence • Assembling and testing electronic circuits • Using multimeter, clamp meter, Megger, function generator, oscilloscope and power analyzer
Underpinning Attitudes	• Commitment to occupational health and safety • Promptness in carrying out activities • Sincere and honest to duties • Environmental concerns • Eagerness to learn • Tidiness and timeliness • Respect for rights of peers and seniors in the workplace • Communication with peers, subordinates and seniors in the workplace

Resource Implications	The following resources must be provided: • Workplace (simulated or actual) • Electrical and electronic circuit trainer kits • Three-phase supply with load bank • Measuring and testing instruments • Machine manuals and drawings • Learning manual and projector

Assessment Evidence Guide:

Critical Aspects of Competency	Assessment requires evidence that the candidate: • interpreted electrical and electronic drawings and documents • analyzed DC and AC circuits • analyzed three-phase systems and power • tested basic electronic components and circuits • used electrical measuring and testing instruments
Methods of Assessment	Competency should be assessed by: • Written test • Practical demonstration • Oral questioning • Portfolio (optional)
Context of Assessment	• Competency assessment must be done in an assessment/training center or in an actual or simulated textile mill workplace after completion of the training. • Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INSTALL AND MAINTAIN INDUSTRIAL POWER DISTRIBUTION SYSTEMS IN TEXTILE MILLS	Nominal Duration: 40 hrs.	Unit Code: SICIP-BPI-IEET-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to install and maintain industrial power distribution systems in textile mills. It specifically includes the tasks of interpreting power distribution requirements, installing LT panels and distribution boards, installing cables, cable trays and busbar trunking, installing earthing and lightning protection, testing and commissioning power distribution installations and performing preventive maintenance of switchgear and panels.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
Interpret power distribution requirements	1. <u>Power distribution layout</u> of a textile mill from substation to machine is interpreted. 2. Connected load, demand and diversity of textile machinery and utilities are calculated. 3. Ratings of transformers, switchgear, cables and protective devices are determined from load calculations. 4. Work instructions, drawings and specifications are interpreted and materials are listed.
Install LT panels and distribution boards	1. <u>Panels and distribution boards</u> are positioned, aligned and securely mounted as per layout. 2. Busbars, incoming and outgoing <u>protective devices</u> and metering are installed and connected. 3. Incoming and outgoing circuits are connected ensuring correct phase, neutral and earth arrangements. 4. Panels are labelled, ferruled and documented as per standards.
Install cables and cable trays	1. Cables and cable trays are selected according to load, route and environment. 2. Cable trays, ladders and supports are installed with proper spacing and earthing. 3. Cables are laid, dressed, glanded and terminated using lugs, ferrules and correct torque.

Elements of Competency	Performance Criteria
Install earthing and lightning protection	1. Earthing system requirements for textile mills are interpreted. 2. Earth electrodes, earth pits and earthing conductors are installed and bonded. 3. Equipment earthing and bonding of machines, panels and cable trays are completed. 4. Earth resistance is measured and compared with permissible limits. 5. Lightning protection and surge protection devices are installed as per design.
Test and commission power distribution installation	1. <u>Pre-commissioning tests</u> are carried out and results are recorded. 2. Protective devices are set as per load and coordination requirements. 3. Installation is energized in sequence following approved switching procedures. 4. Voltage, current, phase sequence and load balance are verified and documented.
Perform preventive maintenance of switchgear and panels	1. Maintenance schedules and checklists are interpreted. 2. Equipment is isolated and secured following lockout/tagout (LOTO) procedures. 3. <u>Maintenance activities</u> on panels, switchgear and busbars are carried out. 4. Thermal scanning and tightness checks are performed and defects are rectified. 5. Maintenance records are completed and abnormal findings are reported.

Range of Variables:

Variables	Range (may include but not limited to)
Power distribution layout	• 33/11 KV incoming and HT switchgear • LBS, VCB, SF6 Circuit breaker • Distribution transformers • Main LT panel/PCC • Sub-distribution boards for machine lines • Capacitor banks/PFI • Generator and ATS/synchronizing panels • Machine-end isolators
Panels and distribution boards	• Main LT panel/PCC • Motor control centre (MCC) • Sub-distribution boards (SDB) • Lighting distribution boards • Utility panels (compressor, humidification, ETP)

Variables	Range (may include but not limited to)
Protective devices	• Air circuit breaker (ACB) • Moulded case circuit breaker (MCCB) • Miniature circuit breaker (MCB) • RCCB/RCBO/ELCB • Fuses and fuse switches • Surge protection devices (SPD) • Protection relays (overcurrent, earth fault)
Pre-commissioning tests	• Visual inspection • Continuity test • Insulation resistance test • Earth resistance test • Polarity and phase sequence test
Maintenance activities	• Cleaning dust and lint from panels • Tightening terminals, bolts and busbar joints • Checking contactor contacts and breakers • Thermal scanning of joints • Testing RCCB/ELCB operation • Checking indicators, meters and cooling fans • Replacing consumables

Curricular Content Guide:

Underpinning Knowledge	• Power distribution structure in textile mills • Load calculation, demand factor and diversity • Selection of transformers, switchgear, cables and protective devices • Panel construction, busbar and termination practices • Cable installation methods and cable tray systems • Earthing, bonding and lightning protection requirements • Testing and commissioning procedures • Preventive maintenance of switchgear and LOTO procedures
Underpinning Skills	• Calculating loads and selecting ratings • Mounting and wiring LT panels and distribution boards • Installing cable trays, laying, glanding and terminating cables • Installing busbar trunking • Installing and testing earthing systems • Performing pre-commissioning tests and energization • Performing preventive maintenance and thermal scanning

Underpinning Attitudes	• Commitment to occupational health and safety • Promptness in carrying out activities • Sincere and honest to duties • Environmental concerns • Eagerness to learn • Tidiness and timeliness • Respect for rights of peers and seniors in the workplace • Communication with peers, subordinates and seniors in the workplace
Resource Implications	The following resources must be provided: • Workplace (simulated or actual) • Main LT panel, MCC and sub-distribution boards • Cables, cable trays, busbar trunking and accessories • Earthing training set and earth resistance tester • Insulation tester, clamp meter, thermal camera • Standard operating procedures and drawings • Learning manual

Assessment Evidence Guide:

Critical Aspects of Competency	Assessment requires evidence that the candidate: • interpreted power distribution requirements • installed LT panels and distribution boards • installed cables, cable trays and busbar trunking • installed earthing and lightning protection • tested and commissioned power distribution installation • performed preventive maintenance of switchgear and panels
Methods of Assessment	Competency should be assessed by: • Written test • Practical demonstration • Oral questioning • Portfolio (optional)
Context of Assessment	• Competency assessment must be done in an assessment/training center or in an actual or simulated textile mill workplace after completion of the training. • Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INSTALL, OPERATE AND MAINTAIN INDUSTRIAL MOTOR CONTROL SYSTEMS	Nominal Duration: 55 hrs.	Unit Code: SICIP-TEX- IEET-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to install and maintain industrial motor control systems used in textile machinery. It specifically includes the tasks of identifying motors used in textile machinery, selecting control and protective devices, installing and wiring motor starters, connecting and testing motors, testing motor control circuits and maintaining motors and starters.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
Identify motors used in textile machinery	1. <u>Types of motors</u> used in textile machinery and utilities are identified. 2. Motor construction, working principle and characteristics are explained. 3. Nameplate data including power, voltage, current, speed, duty, insulation class and IP rating are interpreted. 4. Motor selection for a given textile application is justified.
Select control and protective devices	1. <u>Control devices</u> are selected according to motor rating and application. 2. <u>Protective devices</u> are selected and set according to motor full-load current. 3. Control circuit diagrams of starters are interpreted. 4. Tools, equipment and materials are collected and checked for usability.
Install and wire motor starters	1. Control and protective devices are mounted in the panel as per layout plan. 2. <u>Motor starter circuits</u> are wired according to power and control circuit diagrams. 3. Latching, Interlocking, Timing and Sequential control are implemented. 4. Wiring is ferruled, dressed and labelled as per standards.
Connect and test motors	1. Motor terminal connections (star/delta, dual voltage, single-phase capacitor connections) are made correctly. 2. Motor insulation resistance and winding resistance are tested before energization. 3. Motor is run; direction of rotation is checked and corrected. 4. No-load and load current, speed, vibration and temperature are measured and recorded.

Elements of Competency	Performance Criteria
Test motor control circuits	1. Control circuit is tested for continuity and correct operation before connecting the motor. 2. Starter operation is tested under supervision and common faults are identified and corrected. 3. Overload relay setting and trip are verified. 4. Test results are documented.
Maintain motors and starters	1. Motor maintenance activities are carried out as per schedule. 2. Bearings, cooling fans and terminal boxes are inspected and serviced. 3. Contactor contacts, coils and overload relays are inspected, cleaned and replaced when required. 4. Motor rewinding requirements are identified and motors are dispatched for repair with documentation.

Range of Variables:

Variables	Range (may include but not limited to)
Types of motors	• Three-phase squirrel cage induction motors • Slip ring induction motors • Single-phase capacitor start/run motors • DC motors, BLDC motors • Servo motors and stepper motors • Permanent magnet synchronous motors • Brake motors, torque motors and inverter-duty motors
Control devices	• Magnetic contactors • Push buttons, selector switches, emergency stop • Timers (on-delay, off-delay, star-delta timers) • Counters • Auxiliary relays • Limit switches and proximity switches • Indicator lamps and buzzers
Protective devices	• Thermal overload relays • Electronic overload relays • MCCB/MPCB (motor protection circuit breaker) • Phase failure/phase sequence relays • Single-phasing preventers • Thermistor protection relays
Motor starter circuits	• Direct on-line (DOL) starter • Star-delta starter (manual and automatic) • Forward-reverse starter with interlock • Sequential starting of multiple motors • Jogging/inching control • Two-speed (Dahlander) motor control • Automatic control using float/level/pressure switches

Variables	Range (may include but not limited to)
Common faults	• Contactor not energizing • Motor humming/not starting • Wrong direction of rotation • Overload tripping • Single phasing • Chattering contactor • Timer malfunction • Loose or burnt terminals
Maintenance activities	• Cleaning lint and dust from motor body and fan cover • Lubricating/replacing bearings • Checking insulation resistance • Checking alignment and coupling/belt tension • Measuring vibration and temperature • Tightening terminals • Inspecting contactor contacts and coils

Curricular Content Guide:

Underpinning Knowledge	• Construction, principle and characteristics of AC and DC motors • Motor nameplate interpretation and selection • Starting methods and starter circuits • Control and protective devices and their settings • Motor connection methods • Motor testing procedures • Motor and starter maintenance practices • Common motor and starter faults
Underpinning Skills	• Identifying motors and interpreting nameplate data • Selecting and setting control and protective devices • Wiring DOL, star-delta, forward-reverse and sequential starters • Connecting and testing three-phase and single-phase motors • Testing and fault-finding control circuits • Maintaining motors and starters
Underpinning Attitudes	• Commitment to occupational health and safety • Promptness in carrying out activities • Sincere and honest to duties • Environmental concerns • Eagerness to learn • Tidiness and timeliness • Respect for rights of peers and seniors in the workplace • Communication with peers, subordinates and seniors in the workplace

Resource Implications	The following resources must be provided: • Workplace (simulated or actual) • Three-phase and single-phase motors • Motor starter panels and components • Tools, measuring instruments and testers • Motor manuals and circuit diagrams • Learning manual

Assessment Evidence Guide:

Critical Aspects of Competency	Assessment requires evidence that the candidate: • identified motors used in textile machinery • selected control and protective devices • installed and wired motor starters • connected and tested motors • tested motor control circuits • maintained motors and starters
Methods of Assessment	Competency should be assessed by: • Written test • Practical demonstration • Oral questioning • Portfolio (optional)
Context of Assessment	• Competency assessment must be done in an assessment/training center or in an actual or simulated textile mill workplace after completion of the training. • Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INSTALL, CONFIGURE AND OPERATE VARIABLE FREQUENCY DRIVES (VFD) AND SOFT STARTERS	Nominal Duration: 40 hrs.	Unit Code: SICIP-TE- IEET-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to install, configure and operate variable frequency drives (VFD) and soft starters used in textile machinery. It specifically includes the tasks of interpreting VFD and soft starter principles and applications, installing and wiring VFDs and soft starters, configuring drive parameters, commissioning and operating drives and troubleshooting and maintaining drives.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
Interpret VFD and soft starter principles and applications	1. Working principle of VFD (rectifier, DC bus, inverter, PWM) is explained. 2. Working principle of soft starters is explained. 3. Textile applications of VFDs and soft starters are identified. 4. Drive selection based on motor rating, load type and environment is justified.
Install and wire VFDs and soft starters	1. Drives are mounted in panels with adequate ventilation, clearance and protection from lint. 2. Power wiring, motor wiring and earthing are completed using installation practices recommended by manufacturers. 3. <u>Control wiring</u> for start/stop, speed reference and feedback is completed. 4. Line reactors, chokes, EMC filters and braking resistors are installed where required.
Configure drive parameters	1. Drive keypad/software is used to navigate parameter groups. 2. Motor <u>parameters</u> are entered from the motor nameplate. 3. Application parameters are set as per machine requirements. 4. Parameter settings are backed up and documented.
Commission and operate drives	1. Pre-commissioning checks are carried out before energization. 2. Auto-tuning and test run are performed with the motor uncoupled and then coupled. 3. Speed, acceleration/deceleration and torque performance are verified against machine requirements. 4. Drive is integrated with PLC/HMI or machine controller and operation is verified.

Elements of Competency	Performance Criteria
Troubleshoot and maintain drives	1. Drive fault codes and alarms are interpreted using manuals. 2. Faults in power, motor, control wiring and parameters are diagnosed and rectified. 3. Preventive drive maintenance activities are carried out. 4. Faulty drives are replaced and parameters are restored from backup.

Range of Variables:

Variables	Range (may include but not limited to)
Control wiring	• Digital inputs (start/stop, forward/reverse, multi-speed) • Analog inputs (0–10 V, 4–20 mA potentiometer speed reference) • Relay/digital outputs (run, fault) • Analog outputs (speed, current) • Communication (Modbus RTU, Profibus, Ethernet/IP, Profinet) • Encoder feedback
Parameters	• Motor rated voltage, current, power, speed, frequency, power factor • Control mode (V/f, sensorless vector, closed-loop vector) • Acceleration and deceleration time • Minimum and maximum frequency • Torque boost and torque limit • Overload protection • Braking method • Skip frequencies • PID control • Multi-speed settings
Drive fault codes	• Overcurrent (OC) • Overvoltage (OV) • Undervoltage (UV) • Overload (OL) • Overtemperature (OH) • Earth fault (GF) • Output phase loss • Input phase loss • Communication error • Encoder fault

Variables	Range (may include but not limited to)
Drive maintenance	• Cleaning heat sinks and fans with dry air • Replacing cooling fans • Checking DC bus capacitors • Tightening power terminals • Checking panel filters • Monitoring drive temperature and running data • Firmware/parameter backup

Curricular Content Guide:

Underpinning Knowledge	• Power electronics basics: rectifier, DC bus, IGBT inverter, PWM • V/f and vector control principles • Soft starter principles • Selection of VFDs and soft starters for textile loads • Installation requirements: cables, earthing, EMC, cooling • Parameter structure and commissioning procedure • Drive communication basics • Fault codes and troubleshooting • Drive maintenance requirements
Underpinning Skills	• Mounting and wiring VFDs and soft starters • Configuring motor and application parameters • Performing auto-tuning and test runs • Connecting drives to PLC/HMI • Interpreting fault codes and rectifying faults • Performing preventive maintenance and parameter backup
Underpinning Attitudes	• Commitment to occupational health and safety • Promptness in carrying out activities • Sincere and honest to duties • Environmental concerns • Eagerness to learn • Tidiness and timeliness • Respect for rights of peers and seniors in the workplace • Communication with peers, subordinates and seniors in the workplace
Resource Implications	The following resources must be provided: • Workplace (simulated or actual) • VFDs (0.75–5.5 KW) and soft starters with manuals • Three-phase motors with load • PLC/HMI trainer • Measuring instruments • Learning manual

Assessment Evidence Guide:

Critical Aspects of Competency	Assessment requires evidence that the candidate: • interpreted VFD and soft starter principles and applications • installed and wired VFDs and soft starters • configured drive parameters • commissioned and operated drives • troubleshoot and maintained drives
Methods of Assessment	Competency should be assessed by: • Written test • Practical demonstration • Oral questioning • Portfolio (optional)
Context of Assessment	• Competency assessment must be done in an assessment/training center or in an actual or simulated textile mill workplace after completion of the training. • Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PROGRAM AND OPERATE PLC AND HMI SYSTEMS FOR TEXTILE MACHINERY AUTOMATION	Nominal Duration: 50 hrs.	Unit Code: SICIP-TEX-IEET-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to program and operate programmable logic controllers (PLC) and human machine interfaces (HMI) for textile machinery automation. It specifically includes the tasks of identifying PLC hardware and architecture, wiring PLC inputs and outputs, developing ladder logic programs, configuring HMI, commissioning PLC-based control of textile machine functions and troubleshooting PLC systems and managing program backups.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify PLC hardware and architecture	1.1 <u>PLC components</u> and their functions are identified. 1.2 Types of PLC (compact, modular) and common brands used in textile machinery are recognized. 1.3 PLC scan cycle, memory organization and addressing are explained. 1.4 Programming software is installed and communication with PLC is established.
Wire PLC inputs and outputs	1. <u>Field devices</u> are connected to digital and analog input modules using sink/source configuration. 2. Output devices are connected to relay, transistor and analog output modules. 3. Power supply, earthing and shielding requirements of PLC systems are complied with. 4. I/O wiring is tested and I/O status is verified from the software.
Develop ladder logic programs	1. Ladder logic instructions are used to implement control requirements. 2. Programs for <u>textile machine functions</u> are developed from control descriptions. 3. Programs are documented with tags, comments and I/O lists.
Configure HMI	1. HMI hardware is connected to the PLC and communication is configured. 2. <u>HMI screens</u> are designed with control, monitoring and alarm objects. 3. Tags are linked between HMI and PLC and tested. 4. User levels, recipes and data logging are configured where required.

Elements of Competency	Performance Criteria
Control VFD with PLC & Sensors	1. VFD is interfaced with PLC. 2. VFD motor's speed is controlled with <u>analog sensors.</u> 3. Encoder is interfaced with PLC.
Commission PLC-based control of textile machine functions	1. I/O checks and dry-run tests are performed before connecting to actuators. 2. Machine sequence is tested step by step and interlocks and safety functions are verified. 3. Commissioning records are completed and the system is handed over.
Troubleshoot PLC systems and manage program backups	1. PLC status LEDs and diagnostic buffer are interpreted. 2. <u>PLC faults</u> in power supply, I/O modules, wiring, communication and program logic are diagnosed. 3. Faulty modules are replaced and the system is restored. 4. Program and HMI backups are taken, labelled and stored securely.

Range of Variables:

Variables	Range (may include but not limited to)
PLC components	• CPU • Power supply • Digital input/output modules • Analog input/output modules • Communication modules • Memory card • Programming cable/Ethernet port • Rack/base
Field devices	• Push buttons, selector switches, emergency stop • Limit switches, proximity sensors, photoelectric sensors • Yarn break sensors and stop motions • Temperature (RTD, TC) and pressure sensors • Contactors, solenoid valves, indicator lamps • VFD run/stop and speed reference
Textile machine functions	• Sequential start/stop of machine motors • Stop motion on yarn/sliver break • Doffing sequence control • Length/count measurement and batch stop • Temperature control of dyeing/drying • Level control of tanks • Tension control • Alarm and fault annunciation • Conveyor/material handling sequence

Variables	Range (may include but not limited to)
HMI screens	• Main/overview screen • Manual/auto operation screen • Parameter setting screen • Alarm screen and history • Trend screen • Recipe screen • Maintenance/diagnostic screen
Analog sensors	• Potentiometer • RTD • TC • Pressure sensors
PLC faults	• Power supply failure • CPU stop/fault • I/O module failure • Loose or wrong wiring • Sensor failure • Communication loss • Program logic error • Battery/memory failure

Curricular Content Guide:

1. Underpinning Knowledge	1.1 PLC architecture, scan cycle and addressing 1.2 Input/output wiring and interfacing 1.3 Ladder logic programming and instructions 1.4 Program structure and documentation 1.5 HMI design and tag linking 1.6 Industrial communication protocols 1.7 Commissioning procedures and safety interlocks 1.8 PLC diagnostics, troubleshooting and backup
Underpinning Skills	• Installing and connecting PLC and HMI hardware • Wiring digital and analog I/O • Writing, simulating and testing ladder programs • Designing HMI screens and alarms • Integrating VFDs with PLC • Diagnosing PLC faults and restoring systems • Taking and managing program backups

Underpinning Attitudes	• Commitment to occupational health and safety • Promptness in carrying out activities • Sincere and honest to duties • Environmental concerns • Eagerness to learn • Tidiness and timeliness • Respect for rights of peers and seniors in the workplace • Communication with peers, subordinates and seniors in the workplace
Resource Implications	The following resources must be provided: • Workplace (simulated or actual) • PLC trainers (Siemens S7-1200/Mitsubishi FX/Delta/Omron or equivalent) with software • HMI panels • Sensors, actuators and VFD for integration • Computers with programming software • Learning manual

Assessment Evidence Guide:

Critical Aspects of Competency	Assessment requires evidence that the candidate: • identified PLC hardware and architecture • wired PLC inputs and outputs • developed ladder logic programs • configured HMI • commissioned PLC-based control of textile machine functions • troubleshoot PLC systems and managed program backups
Methods of Assessment	Competency should be assessed by: • Written test • Practical demonstration • Oral questioning • Portfolio (optional)
Context of Assessment	• Competency assessment must be done in an assessment/training center or in an actual or simulated textile mill workplace after completion of the training. • Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: INSTALL, OPERATE AND MAINTAIN SENSORS, INSTRUMENTATION AND ELECTRONIC CONTROL CIRCUITS	Nominal Duration: 40 hrs.	Unit Code: SICIP-TEX-IEET-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to install and maintain sensors, instrumentation and electronic control circuits used in textile machinery. It specifically includes the tasks of identifying sensors and transducers used in textile machinery, installing and calibrating sensors and transmitters, assembling and testing electronic control circuits, maintaining electronic control cards of textile machines and testing instrumentation loops.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
Identify sensors and transducers used in textile machinery	1. <u>Sensors and transducers</u> are identified by type, principle and output signal. 2. Sensor specifications (supply, output, range, response time, IP rating) are interpreted. 3. Suitable sensors for textile sensing applications are selected. 4. Sensor mounting, alignment and wiring requirements are determined.
Install and calibrate sensors and transmitters	1. Sensors and transmitters are mounted and wired as per manufacturer instructions. 2. Two-wire, three-wire and four-wire connections are made correctly. 3. Sensors and transmitters are calibrated against reference instruments. 4. Signal types are checked and scaled at the controller.
Assemble and test electronic control circuits	1. <u>Electronic control circuits</u> are assembled on breadboard/PCB from circuit diagrams. 2. Circuits are tested using multimeter, oscilloscope and signal generator. 3. Circuit faults are identified and rectified. 4. Soldering and desoldering of through-hole and basic SMD components are performed.
Maintain electronic control cards of textile machines	1. <u>Electronic control cards</u> of textile machines are identified and their functions described. 2. Cards are removed, handled and stored using anti-static precautions. 3. Cards are cleaned, inspected and tested; connectors, fuses and simple components are replaced. 4. Cards are replaced, configured and the machine is tested.

Elements of Competency	Performance Criteria
Test instrumentation loops	1. Instrumentation loop drawings are interpreted. 2. Loop continuity, supply and signal are checked from sensor to controller. 3. Loop performance is verified by simulating input and observing controller response. 4. Test results are recorded and deviations are reported.

Range of Variables:

Variables	Range (may include but not limited to)
Sensors and transducers	• Inductive and capacitive proximity sensors • Photoelectric sensors and fibre-optic sensors • Yarn break/stop motion sensors • Encoders and speed sensors • Thermocouples, RTDs and thermistors • Pressure transmitters and pressure switches • Level sensors (float, capacitive, ultrasonic) • Load cells and tension sensors • Humidity sensors • pH and conductivity sensors (dyeing/ETP) • Flow meters
Electronic control circuits	• Regulated DC power supplies (linear and SMPS) • Transistor and MOSFET switching circuits • Relay driver and opto-isolated interface circuits • Timer and oscillator circuits • Comparator and op-amp signal conditioning circuits • SCR/TRIAC phase control (heater control) • Dimmer/Voltage Regulator • Simple microcontroller I/O interfacing
Electronic control cards	• Machine controller/CPU cards • Input/output interface cards • Power supply units • Motor driver cards (servo/stepper) • Display/keypad boards • Sensor amplifier boards • Heater control boards

Curricular Content Guide:

Underpinning Knowledge	• Working principles of sensors, transducers and actuators • Sensor specifications and selection for textile environments • Wiring conventions: 2/3/4-wire, NPN/PNP • Calibration procedures • Analog and digital signal types and scaling • Electronic circuit assembly, soldering and testing • Anti-static handling of electronic cards • Instrumentation loop testing
Underpinning Skills	• Selecting, mounting and wiring sensors • Calibrating sensors and transmitters • Assembling, soldering and testing electronic circuits • Testing and replacing electronic control cards • Testing instrumentation loops
Underpinning Attitudes	• Commitment to occupational health and safety • Promptness in carrying out activities • Sincere and honest to duties • Environmental concerns • Eagerness to learn • Tidiness and timeliness • Respect for rights of peers and seniors in the workplace • Communication with peers, subordinates and seniors in the workplace
Resource Implications	The following resources must be provided: • Workplace (simulated or actual) • Sensor, transducer & actuator kits • Electronic components, breadboards, soldering stations • Oscilloscope, function generator, calibrators • Sample machine control cards • Learning manual

Assessment Evidence Guide:

Critical Aspects of Competency	Assessment requires evidence that the candidate: • identified sensors and transducers used in textile machinery • installed and calibrated sensors and transmitters • assembled and tested electronic control circuits • maintained electronic control cards of textile machines • tested instrumentation loops

Methods of Assessment	Competency should be assessed by: • Written test • Practical demonstration • Oral questioning • Portfolio (optional)
Context of Assessment	• Competency assessment must be done in an assessment/training center or in an actual or simulated textile mill workplace after completion of the training. • Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DIAGNOSE AND TROUBLESHOOT ELECTRICAL AND ELECTRONIC FAULTS IN TEXTILE MACHINERY	Nominal Duration: 50 hrs.	Unit Code: SICIP-TEX- IEET-07-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to diagnose and troubleshoot electrical and electronic faults in textile machinery. It specifically includes the tasks of collecting fault information, applying systematic troubleshooting procedures, diagnosing faults in power, control and electronic circuits, rectifying faults and replacing components, testing and restoring machines to operation and recording and reporting maintenance activities.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
Collect fault information	1. <u>Fault symptoms</u> are gathered from operators, alarms, HMI messages and observation. 2. Machine history, maintenance records and previous breakdowns are reviewed. 3. Machine manuals, circuit diagrams and fault code lists are collected. 4. Safety precautions and LOTO are applied before inspection.
Apply systematic troubleshooting procedures	1. <u>Troubleshooting techniques</u> are applied to narrow down the fault location. 2. Power, control and signal paths are traced using drawings. 3. Measurements are taken at test points and compared with expected values. 4. Root cause of the fault is identified rather than only the symptom.
Diagnose faults in power, control and electronic circuits	1. <u>Power circuit faults</u> are diagnosed using appropriate instruments. 2. <u>Control circuit faults</u> in relays, contactors, timers, counters, sensors and PLC I/O are diagnosed. 3. <u>Electronic faults</u> in drives, power supplies and control cards are diagnosed to replaceable-unit level. 4. Intermittent faults are investigated using monitoring and data logging.

Elements of Competency	Performance Criteria
Rectify faults and replace components	1. Faulty components are replaced with correct specification parts. 2. Repairs are carried out following manufacturer procedures and standards. 3. Parameters and programs are restored after replacement of drives, PLCs or cards. 4. Temporary repairs are avoided or documented for permanent correction.
Test and restore machine to operation	1. Repaired system is tested in stages before full operation. 2. Safety devices, interlocks and emergency stops are verified. 3. Machine is run with production staff and performance is confirmed. 4. Work area is cleaned and tools and spares are accounted for.
Record and report maintenance activities	1. <u>Maintenance records</u> are completed accurately. 2. Spare parts consumption is recorded and reorder requirements are reported. 3. Recurring faults are analyzed and improvement recommendations are made. 4. Unresolved or major faults are escalated with clear documentation.

Range of Variables:

Variables	Range (may include but not limited to)
Fault symptoms	• Machine not starting • Machine stopping frequently/tripping • Motor overheating or noisy • Speed fluctuation or wrong speed • Drive fault alarms • Sensor false triggering • HMI/PLC communication error • Heater not heating • Burning smell, sparking or smoke • Erratic display or reading
Troubleshooting techniques	• Half-split method • Input-to-output signal tracing • Substitution method • Visual, smell and touch inspection • Comparison with identical machine • Use of fault code and diagnostic tools • Six-step troubleshooting procedure

Variables	Range (may include but not limited to)
Power circuit faults	• Blown fuse or tripped breaker • Single phasing • Line to line, double line to earth, triple line to earth • Loose/burnt terminals • Cable insulation failure • Contactor contact wear • Voltage drop and unbalance • Earth fault
Control circuit faults	• Open control circuit • Faulty push button/limit switch • Faulty relay/contactors coil • Timer failure • Sensor misalignment or failure • Wrong PLC I/O signal • Loose connector • Wiring error after repair
Electronic faults	• VFD IGBT/rectifier failure • DC bus capacitor failure • Power supply failure • Control card component failure • Encoder/feedback failure • Communication cable/port failure • Firmware/parameter corruption
Maintenance records	• Breakdown report • Work order/job card • Downtime log • Spare parts issue record • Preventive maintenance checklist • Root cause analysis report • Machine history card/CMMS entry

Curricular Content Guide:

Underpinning Knowledge	• Systematic troubleshooting methodology • Reading machine documentation and fault codes • Power, control and electronic circuit operation • Use of diagnostic instruments and tools • Safe isolation and testing practices • Component replacement and repair standards • Maintenance documentation and root cause analysis • Spare parts management basics

Underpinning Skills	• Collecting and analyzing fault information • Tracing circuits and taking measurements • Diagnosing power, control and electronic faults • Replacing components and restoring parameters/programs • Testing safety functions and restoring machines • Completing maintenance records and reports
Underpinning Attitudes	• Commitment to occupational health and safety • Promptness in carrying out activities • Sincere and honest to duties • Environmental concerns • Eagerness to learn • Tidiness and timeliness • Respect for rights of peers and seniors in the workplace • Communication with peers, subordinates and seniors in the workplace
Resource Implications	The following resources must be provided: • Workplace (simulated or actual) • Textile machine training module or equivalent with inserted faults • Motor control panels, VFDs, PLC/HMI and sensors with fault simulation • Diagnostic instruments (multimeter, tachometer, clamp meter, Megger, oscilloscope, thermal camera) • Machine manuals, drawings and record formats • Learning manual

Assessment Evidence Guide:

Critical Aspects of Competency	Assessment requires evidence that the candidate: • collected fault information • applied systematic troubleshooting procedures • diagnosed faults in power, control and electronic circuits • rectified faults and replaced components • tested and restored machine to operation • recorded and reported maintenance activities
Methods of Assessment	Competency should be assessed by: • Written test • Practical demonstration • Oral questioning • Portfolio (optional)

Context of Assessment	• Competency assessment must be done in an assessment/training center or in an actual or simulated textile mill workplace after completion of the training. • Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IMPLEMENT ENERGY EFFICIENCY AND POWER QUALITY MANAGEMENT	Nominal Duration: 15 hrs.	Unit Code: SICIP-TEX-IEET-08-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to implement energy efficiency and power quality management in textile mills. It specifically includes the tasks of conducting energy audits of textile electrical loads, improving power factor, identifying and mitigating power quality problems, implementing energy conservation measures and monitoring and reporting energy performance.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
Conduct energy audit of textile electrical loads	1. <u>Major energy-consuming loads</u> in the textile mill are identified and categorized. 2. Electrical parameters are measured using power analyzers and energy meters. 3. Specific energy consumption (kWh per kg of yarn/metre of fabric) is calculated. 4. Energy audit findings are documented.
Improve power factor	1. Existing power factor is measured and penalties/losses are calculated. 2. Capacitor bank size is calculated for target power factor. 3. <u>Power factor correction equipment</u> is installed, connected and set. 4. Capacitor bank performance is verified and maintenance is carried out.
Identify and mitigate power quality problems	1. <u>Power quality problems</u> are identified using power quality analyzers. 2. Sources of harmonics such as VFDs, UPS and electronic loads are identified. 3. Mitigation measures are recommended and implemented within limits of responsibility. 4. Results are verified after implementation.
Implement energy conservation measures	1. <u>Energy conservation measures</u> for motors, drives, lighting, compressed air and HVAC are identified. 2. Cost–benefit and payback of measures are estimated. 3. Approved measures are implemented and commissioned. 4. Green energy options such as rooftop solar are identified.

Elements of Competency	Performance Criteria
Monitor and report energy performance	1. Energy monitoring systems and sub-metering are installed or used. 2. Energy data is collected, analyzed and compared with baselines. 3. Energy performance reports are prepared for management.

Range of Variables:

Variables	Range (may include but not limited to)
1. Major energy-consuming loads	• Spinning machine motors (ring frames, carding, blow room) • Loom and knitting machine motors • Air compressors • Humidification and HVAC plants • Dyeing, drying and stenter machines • Boilers and thermal oil heaters • Pumps and blowers • Lighting
2. Power factor correction equipment	• Fixed capacitor banks • Automatic power factor correction (APFC) panels • Detuned capacitor banks with reactors • Capacitor duty contactors • Power factor controllers
3. Power quality problems	• Low power factor • Harmonic distortion (THD) • Voltage sag, swell and flicker • Voltage and current unbalance • Neutral overloading • Transients and surges • Frequency variation
4. Energy conservation measures	5.1 Replacement with IE3/IE4 energy-efficient motors 5.2 VFD application on fans, pumps and compressors 5.3 Compressed air leak reduction and pressure optimization 5.4 LED lighting and daylight utilization 5.5 Timer/sensor-based lighting and HVAC control 5.6 Reducing idle running of machines 5.7 Waste heat recovery 5.8 Rooftop solar PV

Curricular Content Guide:

Underpinning Knowledge	• Energy consumption pattern of textile mills • Energy audit methodology and specific energy consumption • Power factor and capacitor bank sizing • Harmonics and power quality standards (IEEE 519, IEC) • Energy-efficient motors, drives and lighting • Compressed air and HVAC energy saving • Energy monitoring and reporting • Bangladesh energy efficiency policies and buyer sustainability requirements
Underpinning Skills	• Measuring electrical parameters with power analyzers • Calculating specific energy consumption and savings • Sizing and installing capacitor banks • Identifying and mitigating harmonics and unbalance • Implementing energy conservation measures • Preparing energy reports
Underpinning Attitudes	• Commitment to occupational health and safety • Promptness in carrying out activities • Sincere and honest to duties • Environmental concerns • Eagerness to learn • Tidiness and timeliness • Respect for rights of peers and seniors in the workplace • Communication with peers, subordinates and seniors in the workplace
Resource Implications	The following resources must be provided: • Workplace (simulated or actual) • Power quality analyzer and energy meters • APFC panel and capacitor bank • VFD and motor load • Energy audit formats and software • Learning manual

Assessment Evidence Guide:

Critical Aspects of Competency	Assessment requires evidence that the candidate: • conducted energy audit of textile electrical loads • improved power factor • identified and mitigated power quality problems • implemented energy conservation measures • monitored and reported energy performance
Methods of Assessment	Competency should be assessed by: • Written test • Practical demonstration • Oral questioning • Portfolio (optional)

Context of Assessment	• Competency assessment must be done in an assessment/training center or in an actual or simulated textile mill workplace after completion of the training. • Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

-END OF COMPETENCY STANDARD-

WORKSHOP/LAB FACILITY STANDARD**Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):**

- Classroom – 400 sft (37 sqm)
- Workshop/Lab – 1,250 sft (116 sqm) **OR**
- Classroom cum Workshop – 1,550 sft (144 sqm)

Preconditions and Utility Requirements:

- Three-phase 400 V utility supply with adequate capacity (minimum 30 kVA) with an approved metering system
- Adequate illumination and natural/mechanical ventilation; lab area kept free of lint and dust
- Separate clean earth for PLC/electronic trainers and proper equipment earthing throughout the workshop
- Emergency stop/isolation switch, fire extinguishers and first aid kit in the workshop
- Internet connectivity for software updates, video-based learning and manufacturer resources

Major Training Equipment and Training Facilities

The following conditions must be fulfilled:

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.
- The list denotes the minimum training equipment and facility required to effectively conduct training for this course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of the competency standard (CS) provided by SICIP.
- For the operation of the training course on Industrial Electrical and Electronic Technology for Textile Industry, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.
- The institute will also provide all other hand tools and power tools as per CS for 25 trainees, and arrange adequate seating and classroom setup for the 25 trainees.
- The institute is encouraged to establish a partnership with at least one textile mill for industry visits and on-machine practical sessions.

Course Name:	Industrial Electrical and Electronic Technology for Textile Industry
Course Duration:	360 hours (3 months)
Number of Trainees per Batch:	25
Total Trainees:	200 (8 batches)
Entry Qualification:	HSC / Diploma in Engineering or equivalent (Electrical/ Electronics/ Mechatronics/ Power/ Instrumentation/ Mechanical or related)

Sl. No.	Major Equipment and Training Facilities	Required Quantity
Classroom and Instructional Facility		
1	Computer/laptop with PLC, HMI, drive and circuit simulation software	10
2	Multimedia projector with projection screen	01
3	Internet connectivity for video-based and flipped-classroom delivery	01 line
4	Whiteboard with accessories	01
5	Training seating set and instructor desk	25+01
6	Charts/models: electrical symbols, single-line diagram, textile process flow, motor control circuits, safety signs, electric-shock treatment	01 set
Power Distribution, Generation and Utility Systems		
7	Diesel generator set, 3-phase 400/230 V, ≥ 10 kVA with AMF/ATS panel	01
8	Main LT panel 3-phase 400 V (incoming MCCB/ACB, busbar, CTs, voltmeter/ammeter/energy meter, outgoing MCCBs/MCBs, indicator lamps)	01
9	Motor control centre (MCC) demonstration panel with feeders	01
10	Sub-distribution boards (with MCB, MCCB, RCCB/ELCB, SPD)	05
11	Transformer bank, 3 $\times$ single-phase 2–5 kVA, 400/230 V	01 set
12	Automatic power factor correction (APFC) panel with capacitor bank ≥ 10 kVAr and PF controller	01
13	3-phase energy meter (direct and CT-operated) and single-phase energy meter	02 + 02
14	Load bank: resistive/inductive, 3-phase, ~ 5 kW	02
15	Battery bank 12 V/24 V ≥ 50 Ah with charger and battery bench	01 set
16	Busbar trunking system with tap-off boxes (demonstration length)	01 set
17	Cable tray (perforated and ladder) with supports and accessories	02 sets
18	Earthing installation training set (rod, plate/chemical electrode, earth pit with chamber, accessories)	02 sets
19	Lightning protection and surge protection demonstration set	01 set
Motor Control and Drive Systems		
20	3-phase squirrel cage induction motor, 3–5 HP, 400 V (IE3, TEFC)	06

Sl. No.	Major Equipment and Training Facilities	Required Quantity
21	Single-phase motor 0.5–1 HP	04
22	DC motor with controller (demonstration)	01
23	Servo motor with servo drive (demonstration)	01 set
24	Motor starter panels: DOL ×2, star-delta ×2, forward-reverse ×2 (contactor, overload relay, timer, push buttons, selector switch, indicator lamps)	06
25	Magnetic contactors with thermal overload relay (spare/for practice)	25
26	Motor protection circuit breakers (MPCB) and phase failure relays	06 + 04
27	Variable frequency drive (VFD) 0.75–5.5 kW with keypad, manual and PC software	06
28	Soft starter 5.5–7.5 kW	02
29	Line reactor, EMC filter and braking resistor set for VFD	02 sets
30	Encoder/tachometer for speed feedback	02
Automation, Instrumentation and Electronics		
31	PLC trainer (compact PLC with ≥14 DI, 10 DO, 2 AI, 1 AO; e.g., Siemens S7-1200, Mitsubishi FX5, Delta or equivalent) with programming software and cable	10
32	HMI touch panel 7" with software	10
33	Industrial communication kit (Modbus RTU/TCP, Profinet/Ethernet switch, cables)	10 sets
34	Sensor training kit: inductive, capacitive, proximity, photoelectric, fibre-optic, yarn break sensor, limit switch, level switch, pressure switch	10 sets
35	Temperature sensors (RTD Pt100, thermocouple K) with transmitters and temperature controller	10 sets
36	Pressure transmitter (4–20 mA) and level transmitter	10
37	Load cell/tension sensor with amplifier	01 set
38	Process calibrator (mA/V/RTD/TC simulator)	02
39	Electronic circuit trainer (power supply, transistor, op-amp, timer, logic, SCR/TRIAC, IGBT, dimmer, modules)	06
40	Breadboards, component kits and PCB practice boards	13 sets
41	Soldering/desoldering station with anti-static mat and wrist strap	06
42	Digital storage oscilloscope, ≥50 MHz, 2-channel	05
43	Function/signal generator	05

Sl. No.	Major Equipment and Training Facilities	Required Quantity
44	Regulated DC power supply, 0–30 V/5 A	06
45	Textile machine electrical training module (simulated machine with motor, VFD, PLC/HMI, sensors, stop motions and fault-insertion facility) OR access to actual textile machines through industry partnership	01
Workshop Support Facilities		
46	Wiring/panel practice boards with DIN rail, trunking, terminal blocks	12
47	Work bench with bench vice	06
48	Tool storage cabinet	02
49	Material storage rack	02
50	Fire extinguishers (CO2 and dry powder) and first aid kit with CPR chart	02 + 01
51	PPE set (safety glasses, insulated gloves, helmet, dust mask, safety shoes) for trainees	25 sets
52	LOTO kit (locks, tags, hasps)	05 sets
Major Tools, Power Tools and Measuring Instruments		
53	Insulated screwdriver set	25
54	Combination pliers	25
55	Long-nose pliers	13
56	Wire stripper and cutter	25
57	Crimping pliers (ferrule and lug)	13
58	Hydraulic cable lug crimper	02
59	Neon tester/voltage detector	25
60	Digital multimeter (true RMS), energy meter	20
61	Clamp meter (AC/DC, true RMS)	08
62	Insulation resistance tester (Megger) 1000 V	03
63	Earth resistance tester	02
64	Phase sequence tester	03
65	Power quality analyzer (3-phase, harmonic analysis)	02
66	Digital tachometer (contact/non-contact)	03
67	Infrared thermometer/thermal imaging camera	02
68	Vibration meter	01
69	Electric drill and impact driver	05 + 03

Sl. No.	Major Equipment and Training Facilities	Required Quantity
70	Angle grinder	02
71	Hack saw, pipe cutter, conduit bender	05 each
72	Hammer and chisel set	05
73	Spanner/socket set and torque wrench	05 + 02
74	Measuring tape and spirit level	05 each
75	Cable pulling tools and fish tape	05
76	Ladder	02
77	Toolbox	05

Trainer and Assessor Requirements

- Trainer: B.Sc. in Electrical/Electronic/Mechatronics Engineering with minimum 3 years, or Diploma in Engineering with minimum 5 years, of relevant industrial experience, preferably in textile mill maintenance or automation; competency-based training methodology (CBT&A/ToT) certification is preferred.
- Trainer-to-trainee ratio for practical sessions: 1:12 (minimum two trainers/instructors per batch of 25).
- Assessor: Nationally certified assessor or occupation-specific industry expert with minimum 5 years of experience in textile electrical/electronic maintenance.